



## Lesson 6.4 — Introduction to Logarithms

*Big idea:  $\log_b(x)$  is the exponent.  $\log_b(x) = y$  means  $b^y = x$ . Logs and exponentials are inverses.*

### Quick review

**Definition:**  $\log_b(x) = y$  ■  $b^y = x$ . ( $b > 0$ ,  $b \neq 1$ ,  $x > 0$ .)

**Common log:**  $\log(x)$  means  $\log_{10}(x)$ .

**Natural log:**  $\ln(x)$  means  $\log_e(x)$ .

**Domain / range:** domain  $(0, \infty)$ , range ■.

**Identities:**  $\log_b(1) = 0$ ;  $\log_b(b) = 1$ ;  $\log_b(b^x) = x$ ;  $b^{\log_b(x)} = x$ .

### Worked example

**Worked example.** Evaluate  $\log_2(32)$  and  $\log_5(1/25)$ .

$\log_2(32) = 5$  (since  $2^5 = 32$ ).  $\log_5(1/25) = \log_5(5^{-2}) = -2$ .

### Warm-ups

1. Evaluate  $\log_{10}(1000)$ .
2. Evaluate  $\log_3(9)$ .
3. Rewrite  $4^3 = 64$  as a logarithmic equation.

### Practice

1. Evaluate  $\log_2(64)$ .
2. Evaluate  $\ln(e^5)$ .
3. Solve  $\log_5(x) = 3$  for  $x$ .

### Challenge

1. Find  $x$  in  $\log_x(81) = 4$ .
2. Solve  $2 \log_3(x) = \log_3(36)$  without using a calculator.



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## Answer key — Lesson 6.4

*Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.*

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### Warm-ups

1. 3.
2. 2.
3.  $\log_4(64) = 3$ .

### Practice

1. 6.
2. 5.
3.  $x = 5^3 = 125$ .

### Challenge

1.  $x^4 = 81 \Rightarrow x = 3$  (positive base).
2.  $\log_3(x^2) = \log_3(36) \Rightarrow x^2 = 36 \Rightarrow x = 6$  (must be positive for log domain).